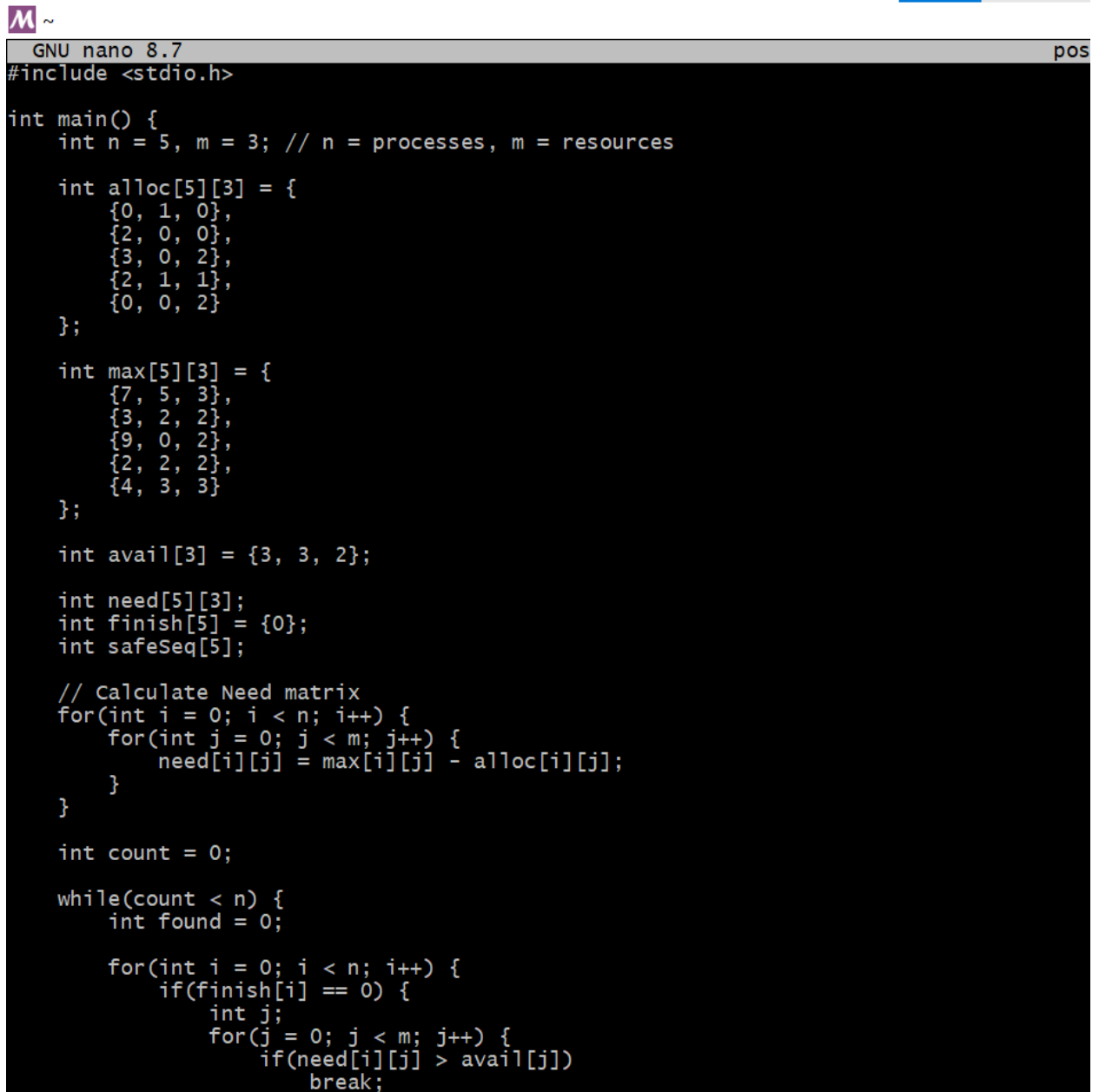


Postlab

Aim: To design and implement the **Banker's Algorithm** for **deadlock avoidance** and to determine whether the system is in a **safe state** or not.

Code:



```
GNU nano 8.7 pos
#include <stdio.h>

int main() {
    int n = 5, m = 3; // n = processes, m = resources

    int alloc[5][3] = {
        {0, 1, 0},
        {2, 0, 0},
        {3, 0, 2},
        {2, 1, 1},
        {0, 0, 2}
    };

    int max[5][3] = {
        {7, 5, 3},
        {3, 2, 2},
        {9, 0, 2},
        {2, 2, 2},
        {4, 3, 3}
    };

    int avail[3] = {3, 3, 2};

    int need[5][3];
    int finish[5] = {0};
    int safeSeq[5];

    // Calculate Need matrix
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++) {
            need[i][j] = max[i][j] - alloc[i][j];
        }
    }

    int count = 0;

    while(count < n) {
        int found = 0;

        for(int i = 0; i < n; i++) {
            if(finish[i] == 0) {
                int j;
                for(j = 0; j < m; j++) {
                    if(need[i][j] > avail[j])
                        break;
                }
            }
        }
    }
}
```

```
GNU nano 8.7 postlab.c

// Calculate Need matrix
for(int i = 0; i < n; i++) {
    for(int j = 0; j < m; j++) {
        need[i][j] = max[i][j] - alloc[i][j];
    }
}

int count = 0;

while(count < n) {
    int found = 0;

    for(int i = 0; i < n; i++) {
        if(finish[i] == 0) {
            int j;
            for(j = 0; j < m; j++) {
                if(need[i][j] > avail[j])
                    break;
            }

            if(j == m) {
                for(int k = 0; k < m; k++)
                    avail[k] += alloc[i][k];

                safeSeq[count++] = i;
                finish[i] = 1;
                found = 1;
            }
        }
    }

    if(found == 0) {
        printf("System is NOT in safe state\n");
        return 0;
    }
}

printf("System is in SAFE state\nSafe sequence: ");
for(int i = 0; i < n; i++)
    printf("P%d ", safeSeq[i]);

return 0;
}
```

Output:



```
Chanchal@DESKTOP-GARLJAI MSYS ~
# nano postlab.c

Chanchal@DESKTOP-GARLJAI MSYS ~
# gcc postlab.c -o postlab

Chanchal@DESKTOP-GARLJAI MSYS ~
# ./postlab
System is in SAFE state
Safe sequence: P1 P3 P4 P0 P2
Chanchal@DESKTOP-GARLJAI MSYS ~
#
```